Preparing Geometry for Overset Meshing Using the Engineering Sketch Pad (ESP)



John F. Dannenhoffer, III

Geocentric Technologies LLC

john@geocentrictech.com

Syracuse University

jfdannen@syr.edu

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Overview

- ESP Overview
  - Objectives
  - Distinguishing Features
- Using ESP to "fix" geometry
  - Adding features
  - Removing features
  - Combining Edges and Faces
- ESP Availability

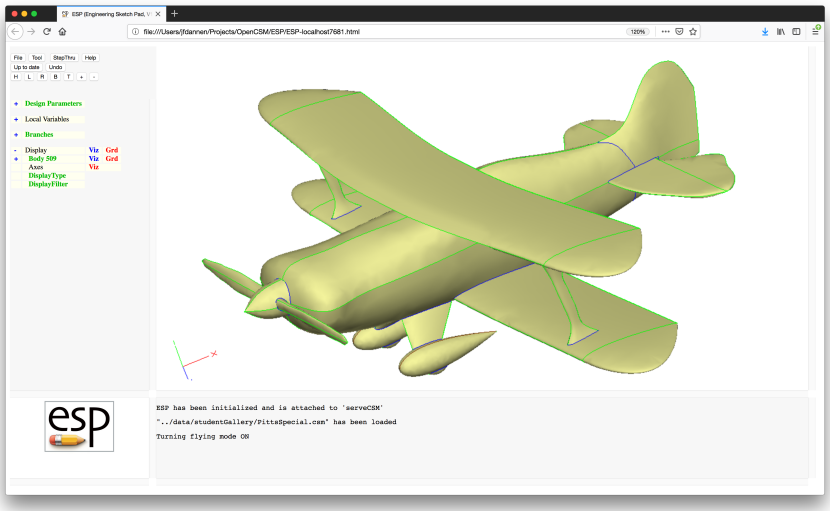
ESP Objectives

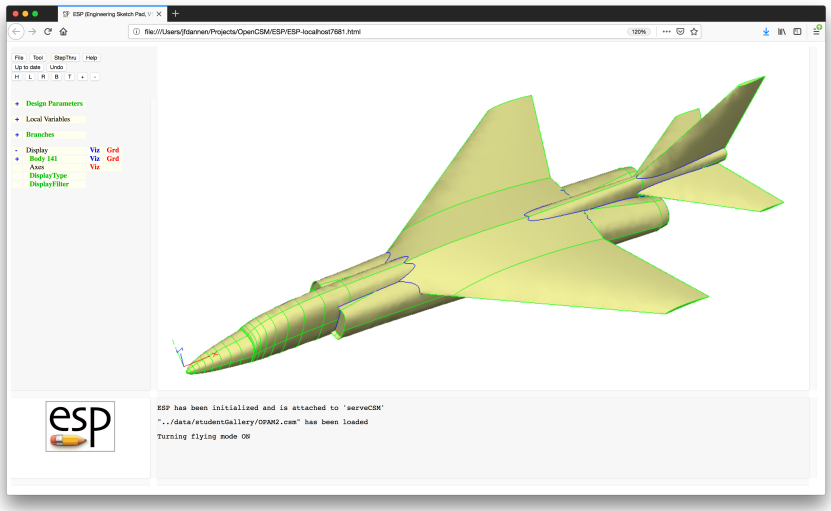
- ESP is a geometry creation and manipulation system specifically designed to support the analysis and design of aerospace vehicles
- It supports multi-fidelity and multi-disciplinary models
- It can be run stand-alone for the development of models
- It can be embedded into other analysis, design, and optimization systems to support their geometry needs

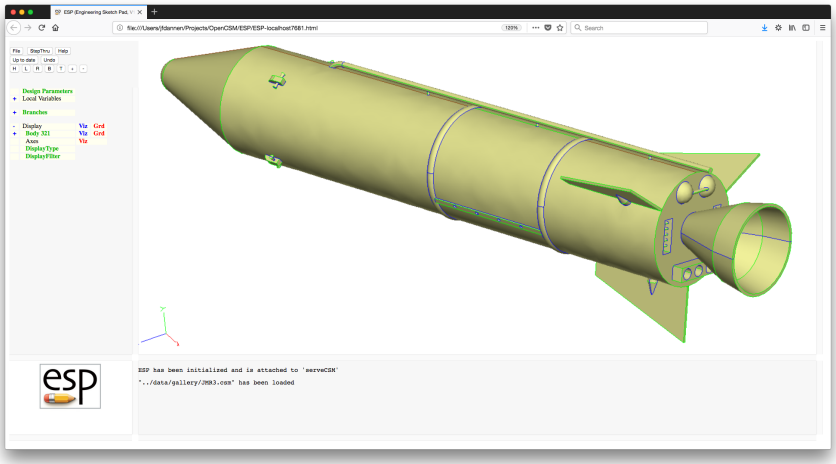
ESP Distinguishing Features

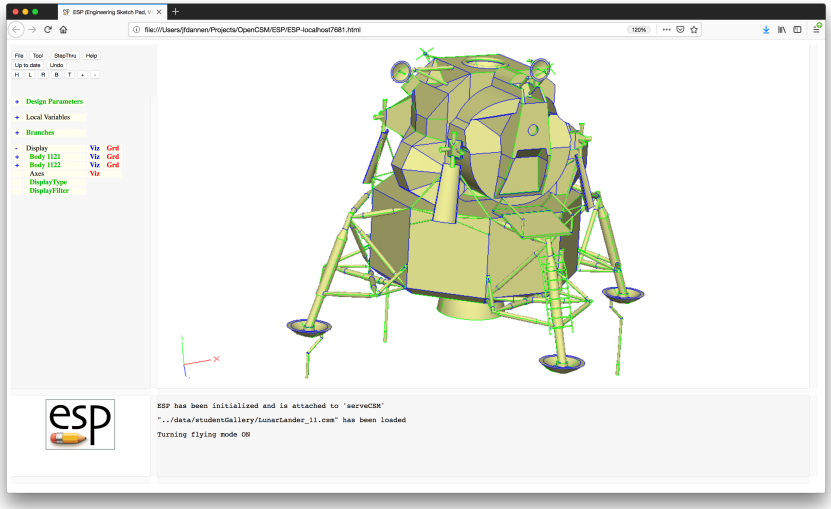
- Solid modeler — constructive solid geometry
- Feature-based — basic and revolved (from sketches) solids and applied features
- Parametric — user-defined, vector-valued parameters
- Associative — persistent attributes across sub-models & builds
- Differentiated — analytic sensitivities (w/ FD backup)
- Extensible — UD-Primitives and UD-Components
- Deployable — back-end server and browser-based client that supports collaboration
- Embeddable — human-readable ASCII `.csm` files
- Open-sourced — freely available under LGPL 2.1

ESP Models Built from Scratch







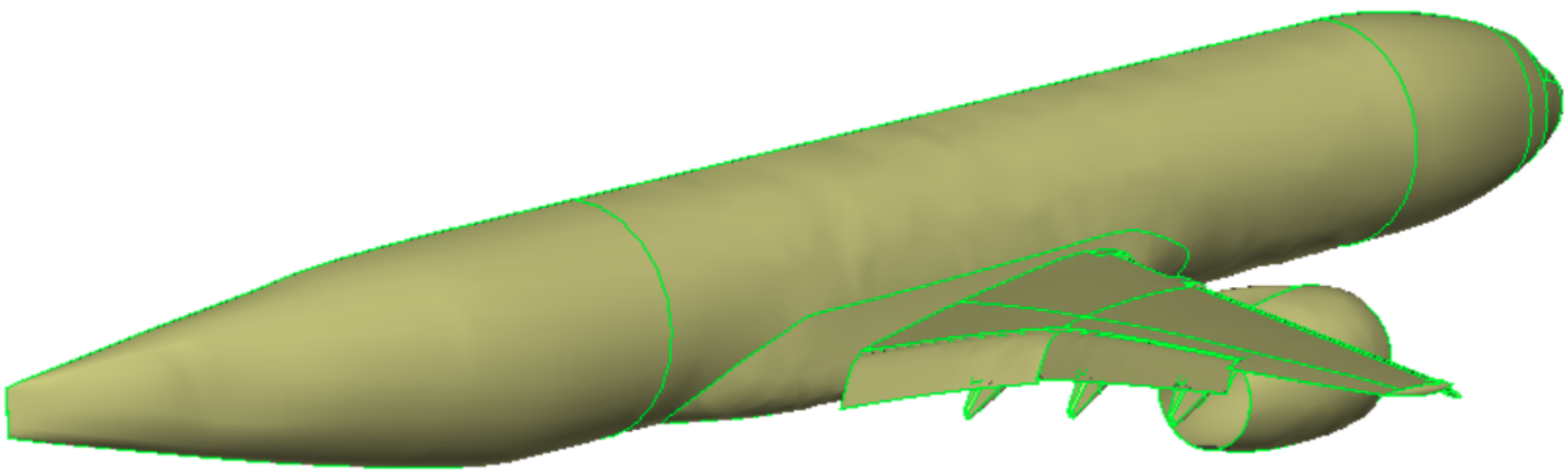


Other ESP-related Capabilities

- `Pyscript` — integrated Python interpreter
- `Miniopt` — non-linear constrained optimizer
- `Vsp3` — interface to OpenVSP models (with sensitivities)
- `Plugs` — parametric geometry from a cloud of points
- `Slugs` — static geometry from a cloud of points
- `EGADS-lite` — light-weight evaluator for HPC environments
- `CAPS` — integration with analyses and optimization

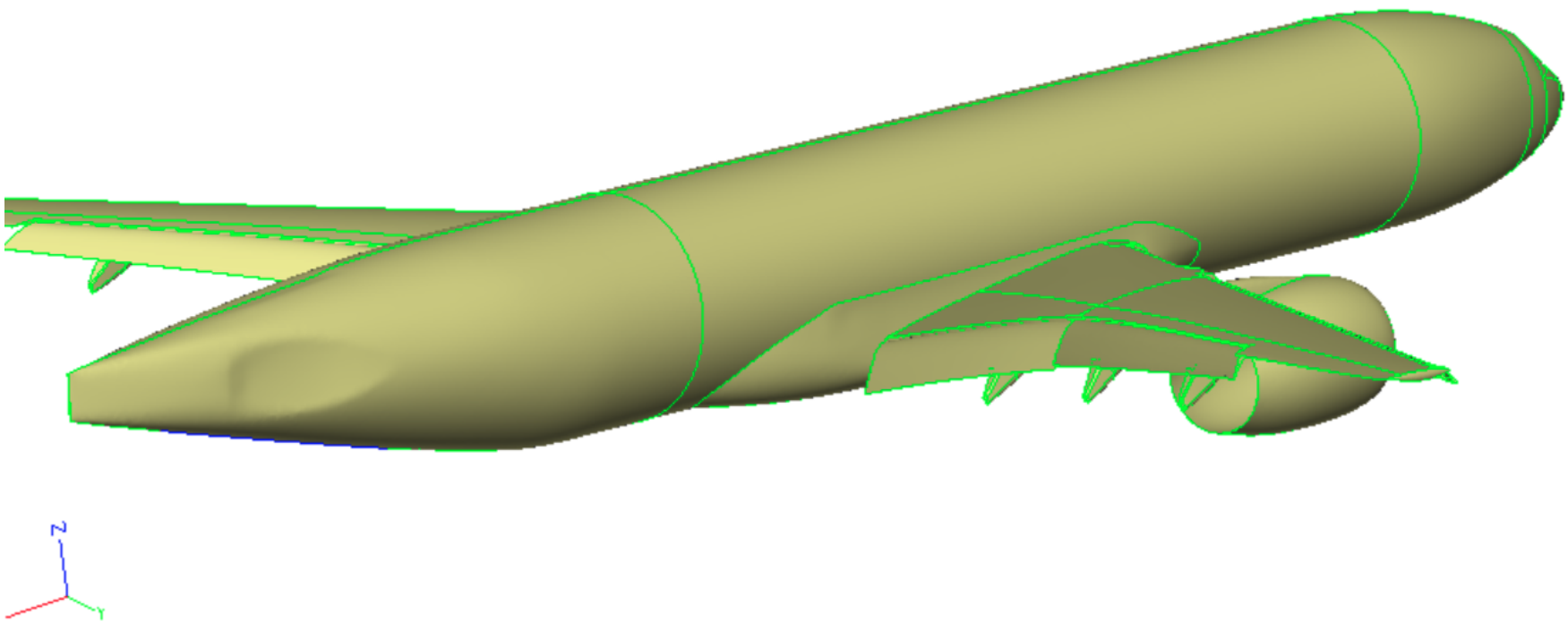
Adding Features — 1

Original configuration (half body in `.stp` file)



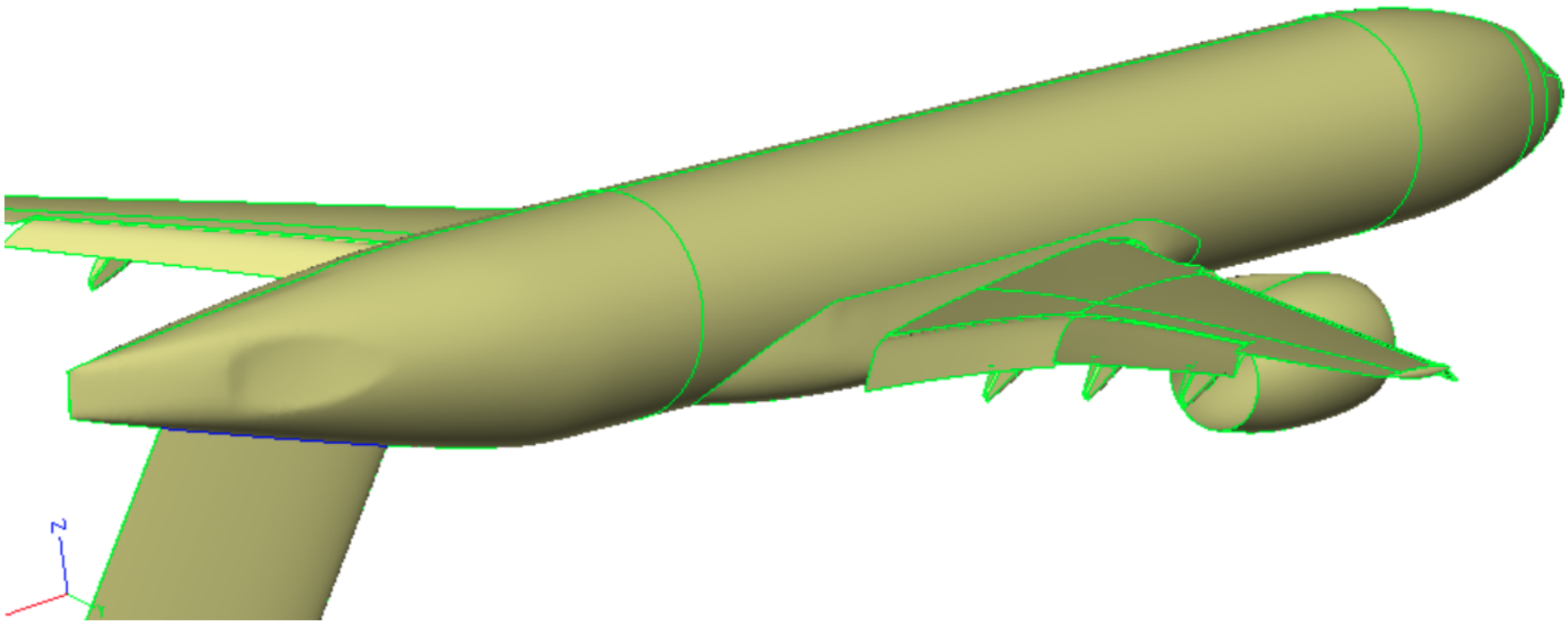
Adding Features — 2

Full body (MIRRORed and JOINed)



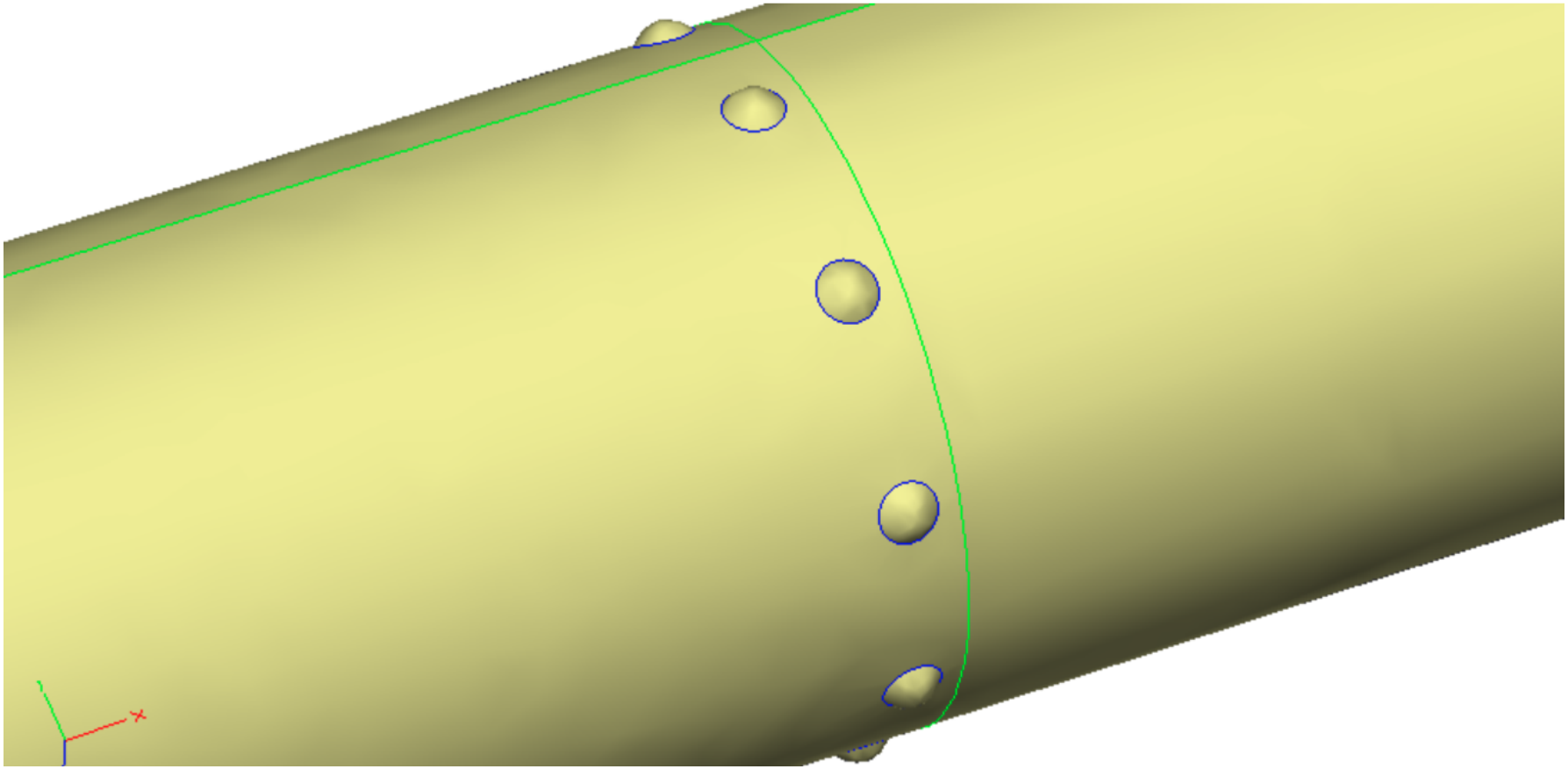
Adding Features — 3

Airfoil-shaped support added



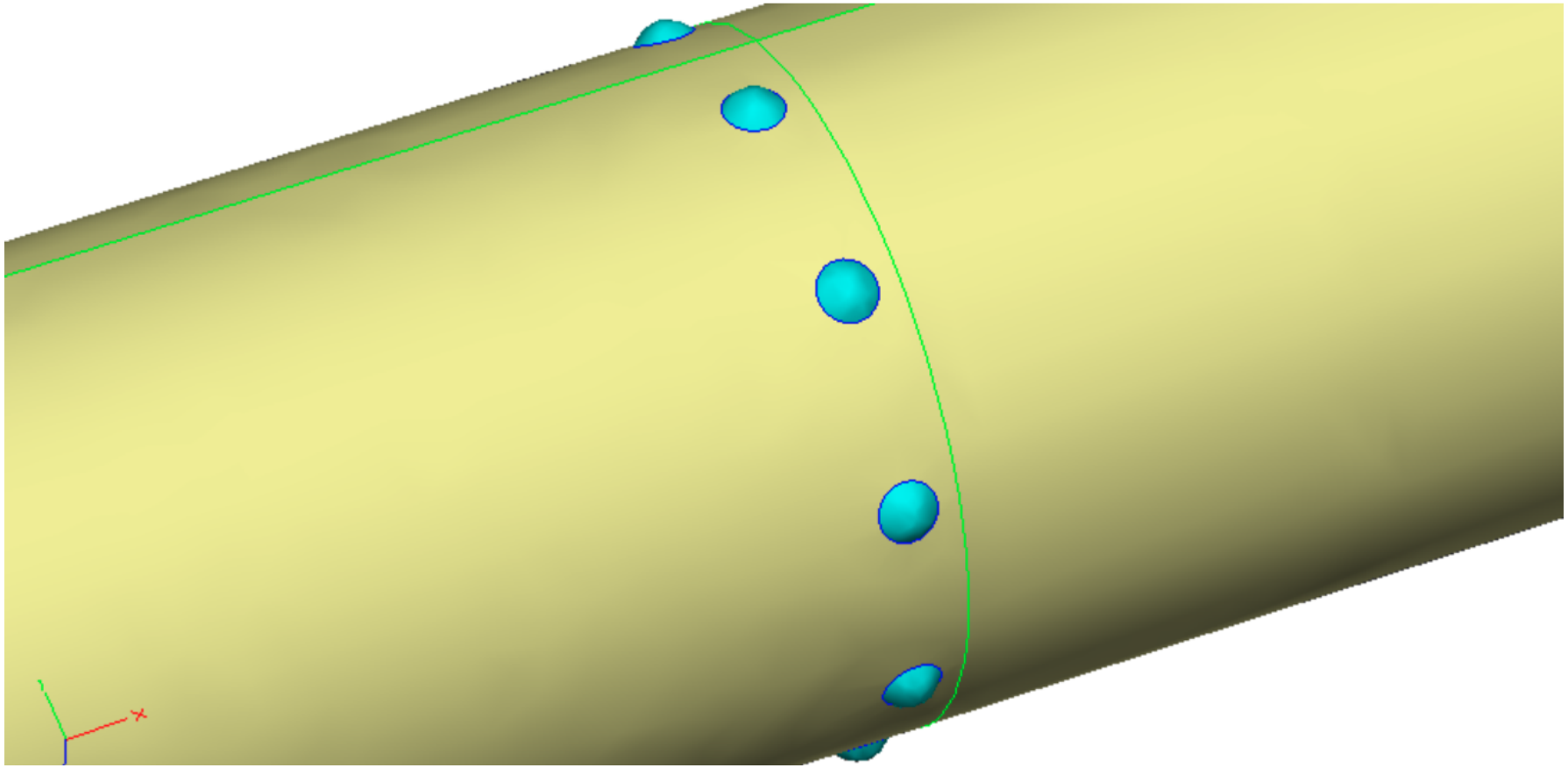
Removing Features — 1

Original configuration



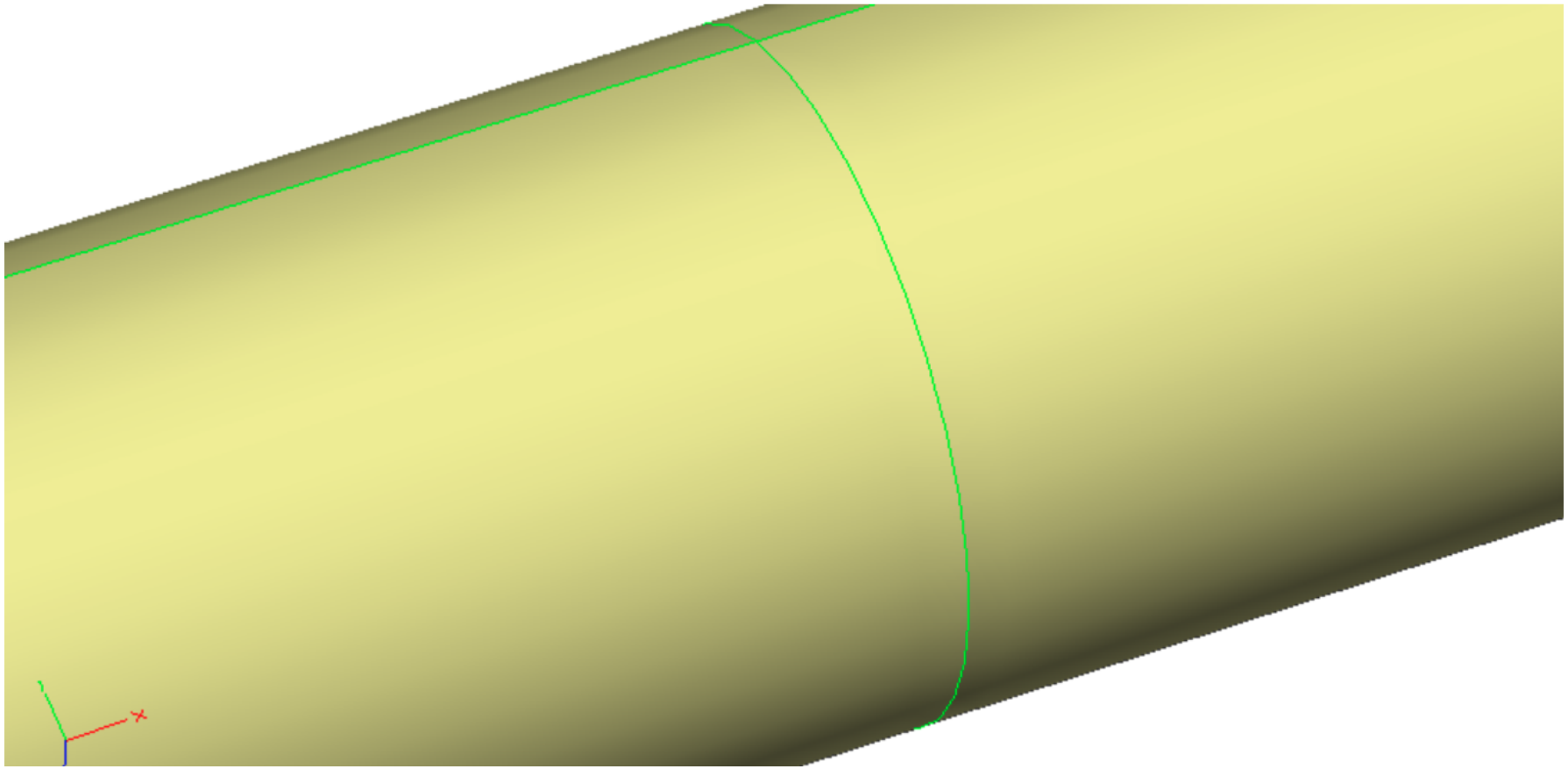
Removing Features — 2

Features to be removed are attributed



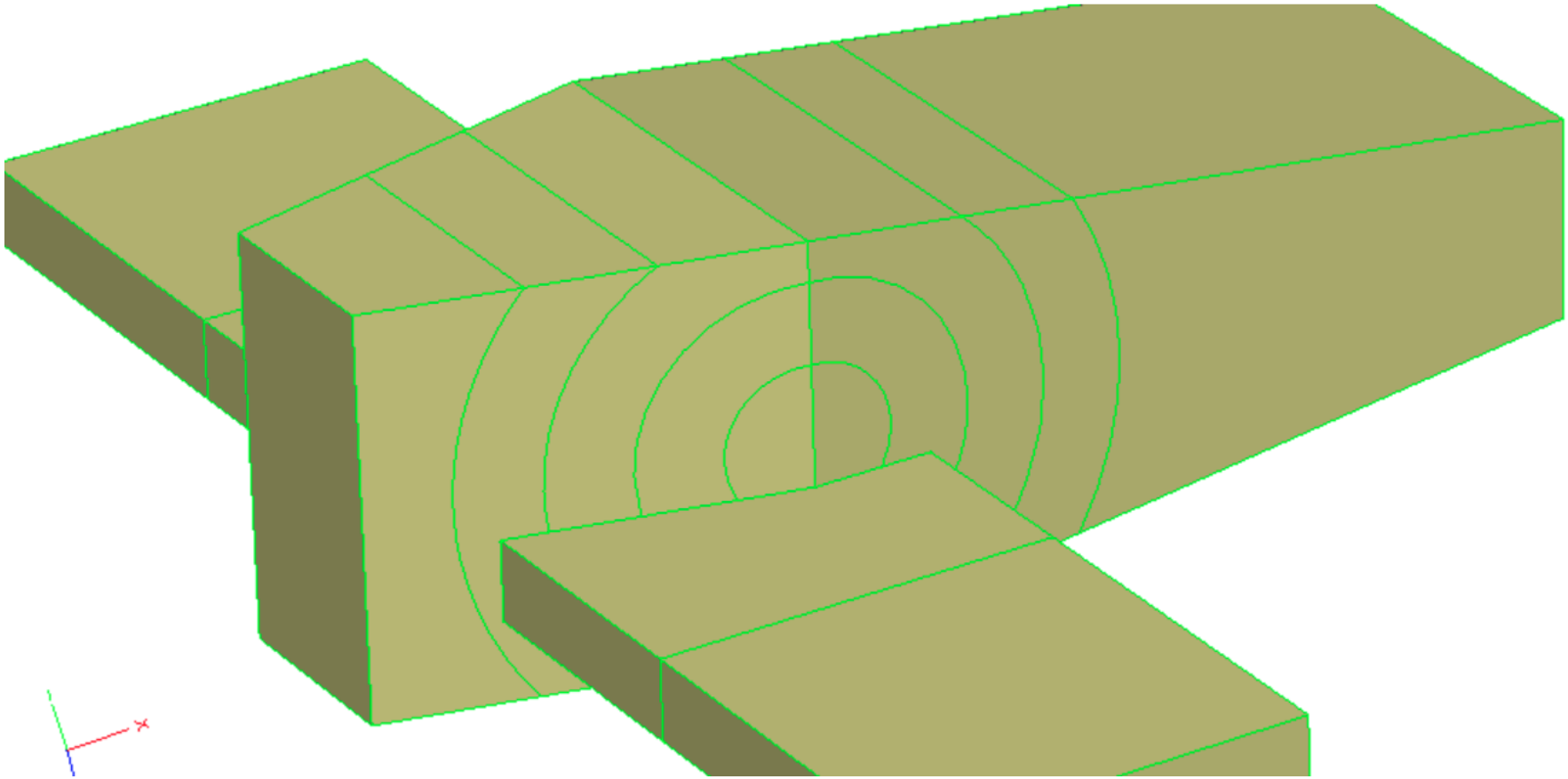
Removing Features — 3

`removeFeatures` UDF invoked



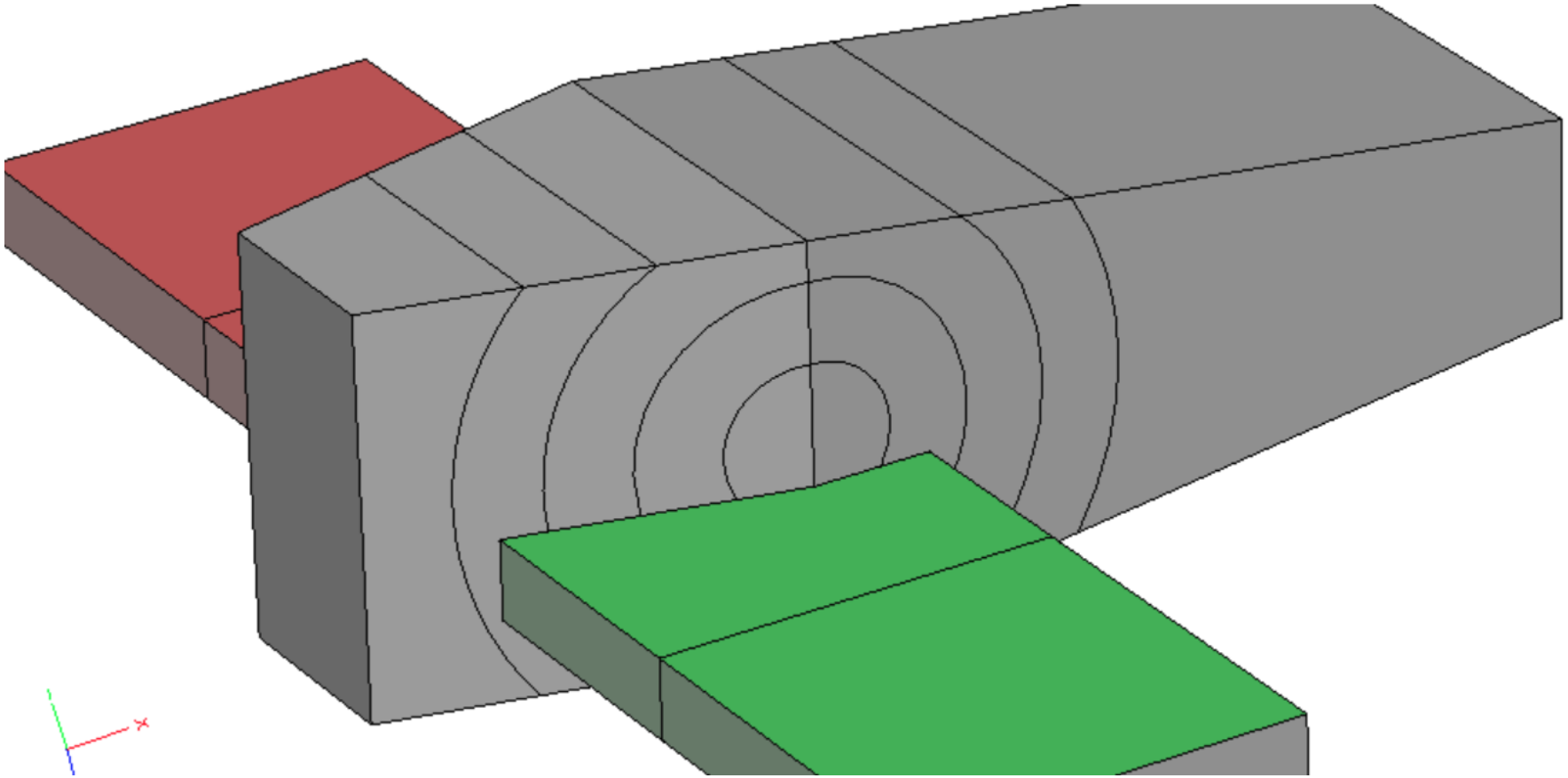
Combining Edges and Faces — 1

Original configuration



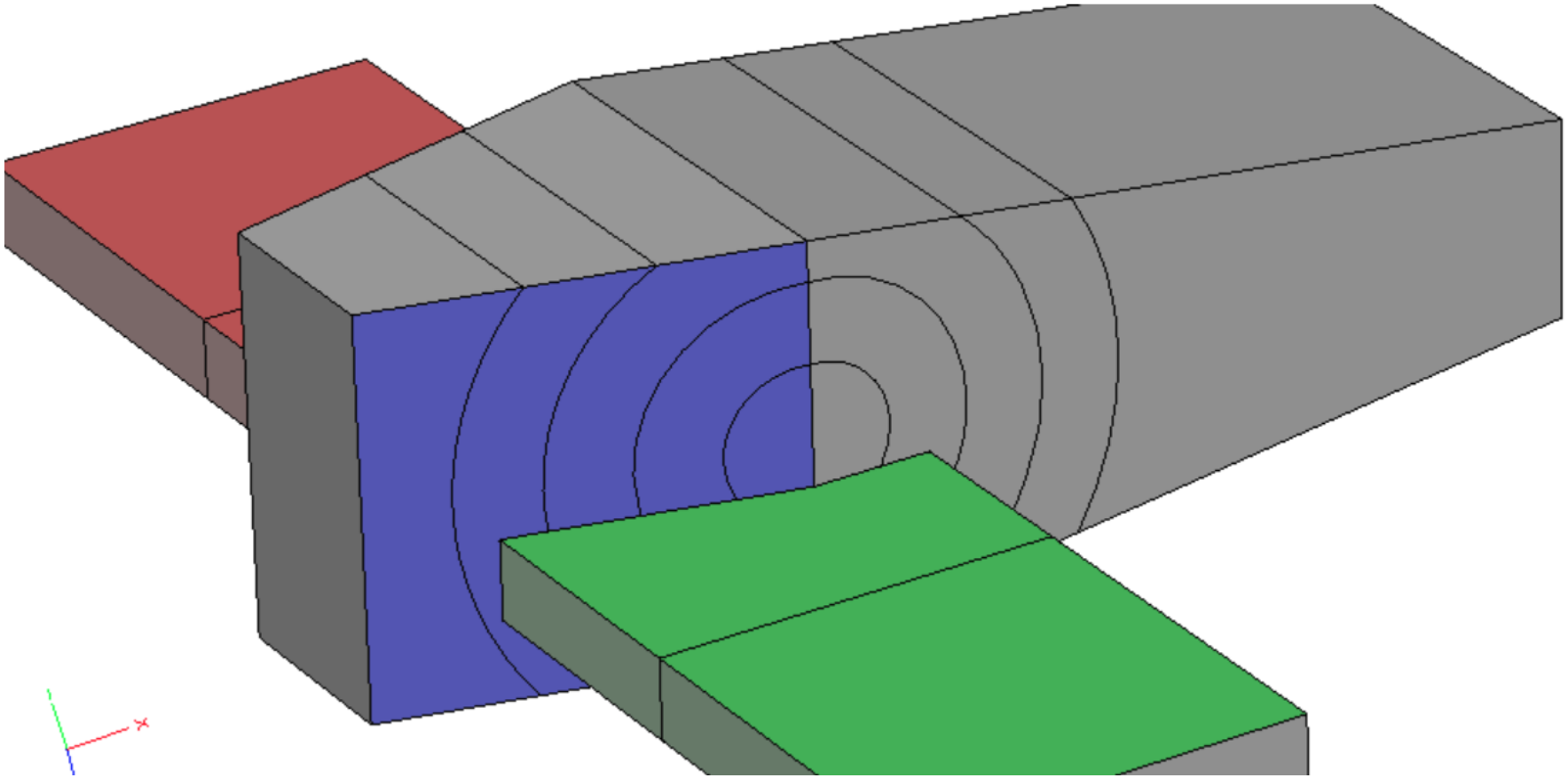
Combining Edges and Faces — 2

Faces colored based upon attribute



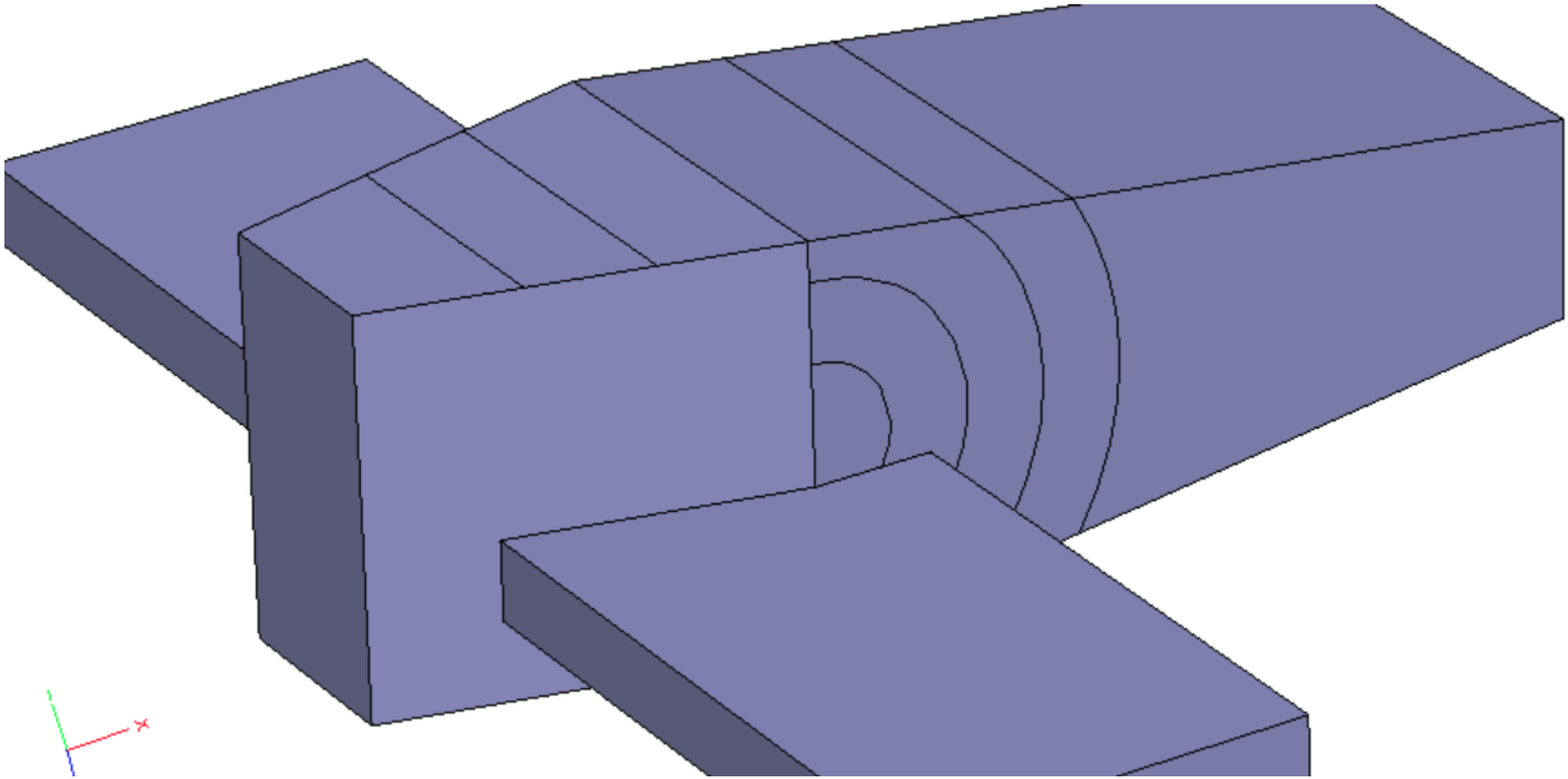
Combining Edges and Faces — 3

Additional Faces colored by mouse-picking



Combining Edges and Faces — 4

Faces combined (and Edges removed)



Combining Edges and Faces — 5

- Result of these operations is an Engineering representation (or Erep)
- Each EFace is a combination of one or more Faces
  - each EFace has its own parametric coordinates $(\bar{u}, \bar{v})$
  - there is a mapping between $(\bar{u}, \bar{v})$ and the (u, v) on the underlying Faces
  - evaluations are actually performed on the underlying Faces (that is, no loss of accuracy)
- Each EEdge is a combination of one or more Edges
  - spurious Nodes are removed
- Each ENode is refers to one Node

Availability

- ESP is free to download and use under LGPL 2.1
- Currently hosted at acdl.mit.edu/ESP
- Will soon be hosted at geocentrictech.com
- Website(s) include:
  - source tarball (zip file)
  - pre-built versions for Linux, MacOS, and Windows
  - training tarball (zip file)
  - ESP publications
  - dependencies: Python and OpenCASCADE
  - archive of all previous versions